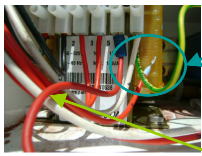
Hard wiring guideline (controller by-pass) for Star Cool standard Reefer unit in case of emergency

The component fitted at the left hand side of the compressor is the frequency converter (FC). It is attached to the compressor with 4 x screws . These screws need to be removed, dismount FC , disconnect power cable and communication cable (power cable still need to be used again for direct power supply to compressor . But, communication cables is not required , ends should be sealed with electrical insulation tape upon disconnection). The FC box might be difficult to dismount due to the locating pins and some calcification between FC and compressor – with the use of a rubber mallet and some friendly persuasion the frequency converter can be dismounted.

1/FC BYPASS

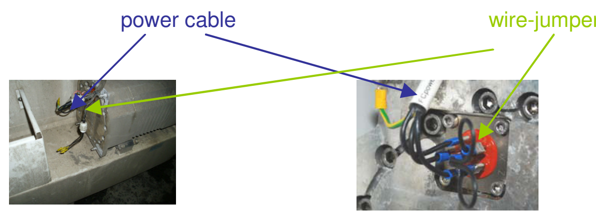
The existing power cable from contactor K8 (L1,L2,L3) to FC need to be connected compressor motor connections as well as a wire-jumper has to be fitted on the remaining 3 terminals. After wiring connections have been done, please access controller, "service menu", access F03--> FC TYPE, the parameter "NONE " to be selected. Then, FC by-pass step completed.

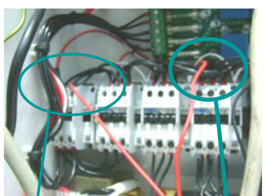


The terminal no. 5 on the transformer output leg (24Vac) to be disconnected

And a new jump wire installed which will be split to power up the other contactors through toggle switch and protection fuse and HPS

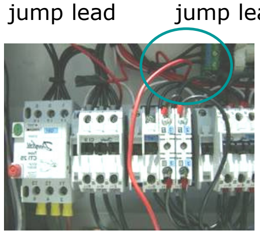
TRAFO T1 approx voltages
Terminal 1-2 : 440V AC
Terminal 3-4 : 20V AC
Terminal 5-6 : 24V AC





K1–main contactor–24v jump lead

K4 – condenser fan (one direction – 24V) jump lead



K7–evaporator fans – 24V jump lead

2/CONTROL CIRCUIT MODIFICATION

The existing 24VAC power supply to the coils of contactors, K1 –K8, at terminal X-35 on interface PCB (wire no.42, 92,90,93,43,94,44,40) need to be removed and wire ends to be terminated with insulation tape. You need 3 jumper

wires to supply control power to contactor coils.

from transformer terminal No.5 (22-24VAC) output to be connected with 3 jumper wires via fuse F6(6.3A), wire no-3 ,

X36 (for safety) for contactor coils. One wire from fuse to contactor coil A1 of contactor K1(CW) or K2 (CCW) to be connected (need to change either K1 or K2 ,depending on the correct running direction of fans/ motors).

Other three jumper leads from fuse for evaporator and condenser contactors and K8(for compressor contactor) to be connected at same point (after fuse).

HP cut out switch connection wires Shp-1 and Shp-2 to be connected in series with K8 control circuit for high pressure protection for compressor.

Evaporator fan motor contactor can be used either K6 or K7(chill range setting to be selected for evap. high speed contactor K7) , K5 for condenser fan high speed, K8 for power supply to FC which is also now used for compressor motor power supply.

Upon power up, check first condenser/evaporator fans running direction. (If evaporator fans were turning the wrong way , change contactor K1 to K2 if necessary– Then, again check for the correct running direction for compressor (use gauge manifold set to monitor for suction and discharge pressures)

3/OPENING OF SOLENOID COILS USING MAGENT

Solenoid coils of **Veco** & **Vexp** to be removed, use permanent magnets (part no.818652A) fitted to open fully or supply coil voltage separately to energize to open.

4/FREON FLOW THROTTLING

After starting the compressor the gas flow should be controlled /throttled manually via valve 14 situated directly above the water cooler. This valve has a 90 degree movement so flow control is very sensitive for chilled cargo

5/ MONITORING

If still possible to supply power(terminal 3 and 4 of transformer=20VAC) to the control panel , you may see the display reading of supply and return temps . But, other parameters such as suction and discharge pressures will not be available . Unit to be monitored closely , use digital thermometer for temperature monitoring as well as pressures to be measured manually by gauge manifold set.

A general view of the 24Vac jumper wiring as well as the main breaker which can be switched on where K1 ; K5 ; K7 and K8 will energize to run the unit

Hard Wiring steps in brief

• 1/frequency converter on the compressor has to be dismounted and bypassed as per guidelines. After wiring done, please access service MENU in controller, select F03 FC TYPE, select “NONE”. By running with FC bypass mode, unit will then run ON-OFF.

• 2/You need to connect power direct from the transformer(24VAC)to the contactors of condenser fan and two evaporator fans and K8(compresor).Please ensure running direction of motors must be checked and correct.

• 3/ TXV eco & TXV exp solenoids need to be removed , open the valves with a permanent magnet (To open the solenoid valves fully, you need to use a permanent magnets , to be fitted on valve plunger each permanently. Alternatively, you can use an external (24VDC supply).

• 4/Also, you need to connect the Hp switch in series with compressor contactor K8 ,or else ,you do not have any safety on the unit (please check HP switch connections removed from terminal no.39 and 89(X35 at interface PCB) and to be connected in series with compressor contactor coil control circuit.

• 5/Then, start the unit, adjust the refrigerant flow with the manually stop valve(valve no.14) placed above the water cooled receiver , you need to install your service gages on the service valve ports in order to monitor the pressures/adjust accordingly.

• 6/Please bear in mind that if you run the unit without the controller then you need to pay much attention to the opening of the manually stop valve (means it must be adjusted more often/ times) because the pressure and temperatures changes all the time.

• 7/ Require to monitor the unit temperatures/pressures manually, if display is not available. Measure the temperature with digital thermo- meter (use YOKOGAWA digital temperature meter or Fluke) ,measure return/supply air as per set point. For frozen cargo, control temperature is return air sensor and supply air sensor to be used as reference for chill cargo .

Kaung 2010 Aug